

Experiment 'p_5_1_a_smallest' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_5_1_a_smallest

Natural language statement: Write a method that returns the smallest of the arguments.

Method signature: p_5_1_a_smallest(x: real, y: real, z: real) returns (smallest: real)

Ensures

- `smallest <= x && smallest <= y && smallest <= z`
- `smallest == x || smallest == y || smallest == z`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the smallest of the arguments.

The signature should be:

`p_5_1_a_smallest(x: real, y: real, z: real) returns (smallest: real)`

The method should respect the following contract:

`ensures smallest <= x && smallest <= y && smallest <= z, ensures smallest == x ||
 ↪ smallest == y || smallest == z`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmyJFLzowiW69xy3ujYM6XaAAmMGE

Model: gpt-5.5-2026-04-23

Created at: 1780563945

Finish reason: stop

Usage: 306 tokens in, and 558 tokens out

```
{  
  if x <= y {  
    if x <= z {  
      smallest := x;  
    } else {  
      smallest := z;  
    }  
  } else {  
    if y <= z {  
      smallest := y;  
    } else {  
      smallest := z;  
    }  
  }  
}
```

Final Program

```
method p_5_1_a_smallest(x: real, y: real, z: real) returns (smallest: real)
  ensures smallest <= x && smallest <= y && smallest <= z
  ensures smallest == x || smallest == y || smallest == z
{
  if x <= y {
    if x <= z {
      smallest := x;
    } else {
      smallest := z;
    }
  } else {
    if y <= z {
      smallest := y;
    } else {
      smallest := z;
    }
  }
}
```

Total Token Usage

Input tokens: 306

Output tokens: 558

Reasoning tokens: 469

Sum of ‘total tokens’: 864

Experiment Timings

Overall Experiment started at 1780563930267, ended at 1780564016856, lasting 86589ms (86.59 seconds)

Iteration #1 started at 1780563930277, ended at 1780564016755, lasting 86478ms (86.48 seconds)